Disc Plough



Category: **Agriculture & Gardening** Type: **Full SOP** Priority: **High** 1 labs

Covers these lab items: Disc plough

What it is used for

A disc plough cuts, lifts, and turns soil using rotating concave discs to prepare land for planting. Correct hitching, disc adjustment, and controlled tractor operation help achieve uniform tillage while reducing equipment damage and soil compaction.

Main parts

- Concave plough discs
- Disc hubs, bearings, and scrapers
- Main frame and disc gangs
- Three-point hitch or drawbar connection
- Depth and angle adjustment mechanism
- Safety reflectors, fasteners, and transport locks where fitted

Before you start

1. Read the operator manual and confirm that the plough is compatible with the tractor's power, hitch, and lifting capacity.
2. Inspect discs for cracks, excessive wear, or deformation and check hubs, bearings, scrapers, and fasteners.
3. Check the frame, hitch points, hydraulic connections, and transport locks for damage.
4. Park the tractor and plough on firm, level ground before hitching or adjustment.
5. Confirm that guards, reflectors, and safety devices are fitted where provided.
6. Survey the field for rocks, buried obstacles, steep slopes, overhead hazards, and people or animals.

How to use

1. Position the tractor safely and attach the plough using the approved hitching procedure.
2. Secure all hitch pins and retainers, and connect hydraulic lines if fitted according to the manual.
3. Adjust the top link, disc angle, working width, and depth settings for the soil and tillage requirement.
4. Raise the plough and verify transport clearance before moving to the field.
5. Travel at a controlled speed and lower the plough gradually at the start of the furrow.
6. Engage the plough at the recommended operating speed and maintain a steady tractor path.
7. Monitor disc penetration, soil turning, tractor load, and the uniformity of the furrow.
8. Adjust depth or angle only according to the manual and safe operating procedure if tillage is uneven.

9. Raise the plough before turning at headlands or crossing obstacles where required.
10. Stop on firm, level ground before inspecting or clearing soil and crop residue from the discs.
11. Lower the implement to the ground, stop the tractor, apply the parking brake, and isolate the engine before adjustments or maintenance.
12. At the end of the job, clean soil from the discs and inspect the implement for damage.
13. Park the plough securely with the implement lowered or supported as specified by the manufacturer.

⚠ Safety

- Only trained operators should operate the tractor and plough combination.
- Keep people and animals clear of the implement, especially during hitching, turning, and raising or lowering.
- Never work beneath a raised plough unless it is secured with the manufacturer's approved support.
- Stop the tractor and isolate the engine before clearing blockages or servicing the discs.
- Use the correct hitch pins and retainers; never rely on improvised connections.
- Reduce speed on slopes and uneven ground, and avoid conditions that may cause rollover or loss of traction.

Care & storage

- Clean soil and crop residue from discs, scrapers, and frame after use.
- Inspect discs, hubs, bearings, and fasteners regularly for wear or damage.
- Lubricate designated points and maintain bearings according to the manual.
- Check hydraulic hoses, hitch points, and transport locks before each use.
- Replace worn or damaged discs and arrange scheduled servicing before the next season.

Common mistakes to avoid

- Using a plough that exceeds the tractor's rated capacity.
- Setting incorrect disc angle or working depth for the soil conditions.
- Turning sharply with the plough engaged in the soil.
- Inspecting or clearing the implement while it is raised or the tractor is running.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.